Project: Worked examples
Title: Pipe pressure drop and pump duty
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Friction and fitting losses through series pipe segments at a given flow, the static head, the system curve, the duty point against a pump curve entered as points, the affinity-law shift for a speed or impeller change, and the NPSH available at the pump suction.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Design flow from the heat carried, when a heat load is given in place of a flow | $Q = \Phi/(\rho\, c_p\, \Delta T)$ |
| Mean velocity in the pipe | $v = Q/(\pi D^2/4)$ |
| Reynolds number, which decides laminar or turbulent flow | $Re = \rho v D/\mu$ |
| Friction factor for laminar flow (Re < 2300) | $f = 64/Re$ |
| Friction factor for turbulent flow, Colebrook–White by fixed-point iteration (at most 100 steps) from the Swamee–Jain value, which also cross-checks it | $1/\sqrt{f} = -2\log_{10}(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$ |
| Head lost in the pipe and its fittings | $h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\, v^2/2g$ |
| System curve, with the friction of a turbulent or transitional segment and every fitting loss scaled from the design flow as $Q^2$, and the friction of a laminar segment (Re < 2300, $h_f \propto Q$) as Q (a closed circuit has $\Sigma dz = 0$, the rise in the flow being recovered in the return) | $H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_{f,lam}\cdot(Q/Q_o) + \Sigma h_{f,other}\cdot(Q/Q_o)^2$ |
| Water held in the pipes | $V_{pipe} = \Sigma\, \pi/4 \cdot D^2 \cdot L$ |
| Duty point, where the pump and system curves cross (every crossing is found by a scan and refined by bisection; the stable one, where the pump curve falls through the system curve, nearest the design flow is reported) | $H_{pump}(Q) = H_{sys}(Q)$ |
| Pump affinity laws for a change of speed n | $Q \propto n$; $H \propto n^2$. For a change of impeller diameter D, by the choice: a trimmed impeller of the same casing, $Q \propto D$ and $H \propto D^2$ (the default, the usual practice for a trim); or a geometrically similar pump scaled in all dimensions, $Q \propto D^3$ and $H \propto D^2$. Speed and diameter combine as $Q \propto n\cdot D\text{^}a$, $H \propto n^2\cdot D^2$, a = 1 or 3 |

| WHAT IT CALCULATES | RELATION |
|---|---|
| Net positive suction head available, to compare with what the pump requires | $NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$, with $p_v$ from the water correlation or, for a custom fluid, from the vapour pressure entered. The friction $h_{f,suction}$ is that of the segments marked 'suction' only, at the larger of the design flow and the duty flow the pump runs at |
| Shaft torque for the motor | $T = P/\omega$ with P the larger of the design-flow shaft power and the duty-point shaft power, at the speed $n_{pump}$; when a speed or diameter ratio shifts the duty, the shifted duty power at the shifted speed $n_{pump} \cdot n_2/n_1$ |

Basis: Flow of fluids through valves, fittings and pipe[1]; Internal flow: friction factor, minor losses, pump duty[2](Ch. 8); Flow of fluids in pipes and ducts (principles)[3](Section 4); Density of liquid water[4].

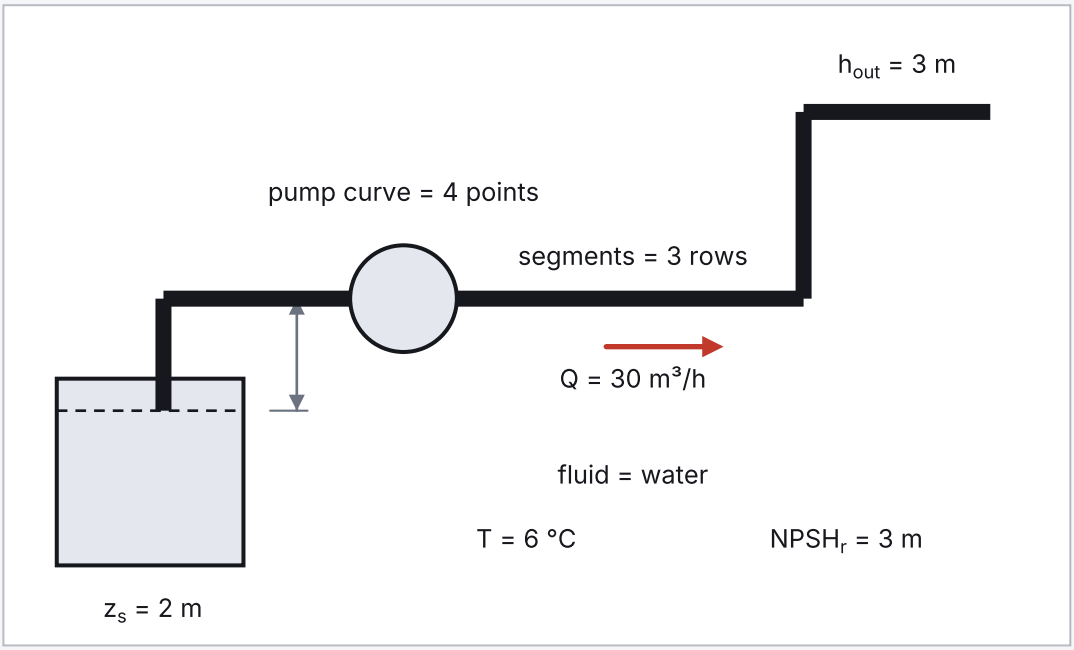



Figure 1 — A pump lifting from a suction level through the pipe segments as listed (three drawn) to an outlet that needs a pressure head. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| fluid | Fluid | water | | spec: SPEC-M-04 §3: chilled water, no glycol | any |
| T | Fluid temperature (mean; link B6 $t_m$ for a heating loop) | 6 | °C | spec: SPEC-M-04 §3: 6 °C flow | 0 … 100 °C |
| Q | Design flow rate (blank when a heat load gives it) | 30 | m³/h | spec: Schedule CHW-01 design flow | 0.01 … 100000 m³/h |
| segments | Pipe segments in series | 3 row(s) | | drawing: DRG-M-110 rev C | any |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| circuit | Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored) | open (the pump lifts the water by the rises entered) | | default | any |
| $h_{out}$ | Pressure head required at the outlet | 3 | m | spec: Terminal unit: 30 kPa ≈ 3 m | 0 … 1000 m |
| $v_{max}$ | Velocity limit (warning) | 2.5 | m/s | spec: CIBSE Guide C practice limit, assumed | 0.5 … 10 m/s |
| $Q_{pump}$ | Pump curve: flow points | [0, 20, 40, 60] | m³/h | spec: Pump datasheet P-01 rev 2, curve points | any |
| $H_{pump}$ | Pump curve: head points | [40, 36, 24, 4] | m | spec: Pump datasheet P-01 rev 2, curve points | any |
| $n_{ratio}$ | Speed ratio $n_2/n_1$ (affinity) | 0.9 | | assumption: Inverter turn-down to 90 % speed. | 0.2 … 2 |
| $D_{ratio}$ | Impeller diameter ratio $D_2/D_1$ (affinity; the law is chosen below) | 1 | | assumption: Impeller unchanged. | 0.7 … 1.1 |
| $D_{law}$ | Law for the diameter ratio: a trimmed impeller of the same casing (Q ∝ D, H ∝ D²) or a geometrically similar pump (Q ∝ D³, H ∝ D²) | impeller trim (Q ∝ n·D, H ∝ n²·D²) | | default | any |
| $\eta_{pump}$ | Pump efficiency at duty (for shaft power) | 0.68 | | spec: Pump datasheet P-01 rev 2 | 0.2 … 0.95 |
| $p_{atm}$ | Atmospheric pressure (NPSH) | 101.325 | kPa | assumption: Sea level. | 50 … 110 kPa |
| $z_s$ | Static head at the pump suction (+ flooded, − lift) | 2 | m | drawing: DRG-M-110 rev C | -10 … 100 m |
| $NPSH_r$ | NPSH required by the pump, at the flow $NPSH_a$ is evaluated at: the larger of the design flow and the duty flow (optional) | 3 | m | spec: Pump datasheet P-01 rev 2 | 0 … 50 m |

| TABLE ROW | VALUES |
|---|---|
| segments[1] | name = plantroom suction; L = 8 m; D = 100 mm; eps = 0.045 mm; K = 3; L_eq = 0 m; dz = 0 m; side = suction |

| TABLE ROW | VALUES |
|---|---|
| segments[2] | name = riser DN80; L = 60 m; D = 80 mm; eps = 0.045 mm; K = 8; L_eq = 0 m; dz = 10 m; side = discharge |
| segments[3] | name = index run DN65; L = 45 m; D = 65 mm; eps = 0.045 mm; K = 12; L_eq = 6 m; dz = 2 m; side = discharge |

## 3. Assumptions

A1. Steady, incompressible, fully developed flow; all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method); both may be used on one segment. (Crane TP-410 practice.)
A3. The system curve scales the friction of turbulent and transitional segments and all fitting losses with $Q^2$ from the design flow (the friction factor is held at its design-flow value); the friction of a laminar segment scales with Q; the static part is independent of flow. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A4. Water properties are from published correlations at atmospheric pressure (0–100 °C); any other fluid is entered with its own density and viscosity and their provenance. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A5. $n_{ratio}$ (Speed ratio $n_2/n_1$ (affinity)) is taken as 0.9. (Inverter turn-down to 90 % speed.)
A6. $D_{ratio}$ (Impeller diameter ratio $D_2/D_1$ (affinity; the law is chosen below)) is taken as 1. (Impeller unchanged.)
A7. $p_{atm}$ (Atmospheric pressure (NPSH)) is taken as 101.325 kPa. (Sea level.)

## 4. Working

[3] ρ = Kell (1975) density of water at T [4]
= T = 6 °C
= 999.94 kg/m³
Density

[4] μ = Vogel fit μ = 0.02939 exp(507.88/($T_K$ - 149.3)) mPa·s
= $T_K$ = 279.15 K
= 1.4684 mPa·s
Dynamic viscosity

[5] $p_v$ = Hyland–Wexler saturation pressure
= T = 6 °C
= 0.93525 kPa
Vapour pressure (NPSH)

[6] $T_{line}$ = T (no $T_{line}$ entered)
= 6 °C
= 6 degC
Design temperature of the line to insulate: the fluid temperature T, since no $T_{line}$ was entered

[10] $Q_{design}$ = Q
= 30 m³/h
= 30 m³/h
Design flow (as entered)

[11] $v_{s1}$ = plantroom suction: Q / (π D²/4)
= 30 m³/h / (π × 100² mm² / 4)
= 1.061 m/s
Velocity, plantroom suction

[12] $Re_{s1}$ = plantroom suction: ρ v D / μ
= 999.94 × 1.061 × 0.1 / 0.0014684
= 72253
Reynolds number, plantroom suction

[13] $f_{s1}$ = plantroom suction: Darcy friction factor
= Colebrook–White, 10 iterations (Swamee–Jain start 0.021152)
= 0.021093
Friction factor, plantroom suction
[14] $f_{sj,s1}$ = plantroom suction: Swamee–Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.021152
Explicit cross-check, plantroom suction
[15] $h_{f,s1}$ = plantroom suction: f (L + $L_{eq}$)/D · v²/2g + K v²/2g
= 0.021093 × (8 + 0) / 0.1 × 0.057399 + 3 × 0.057399
= 0.26906 m
Head loss, plantroom suction
[16] $v_{s2}$ = riser DN80: Q / (π D²/4)
= 30 m³/h / (π × 80² mm² / 4)
= 1.6579 m/s
Velocity, riser DN80
[17] $Re_{s2}$ = riser DN80: ρ v D / μ
= 999.94 × 1.6579 × 0.08 / 0.0014684
= 90316
Reynolds number, riser DN80
[18] $f_{s2}$ = riser DN80: Darcy friction factor
= Colebrook–White, 10 iterations (Swamee–Jain start 0.020946)
= 0.020849
Friction factor, riser DN80
[19] $f_{sj,s2}$ = riser DN80: Swamee–Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.020946
Explicit cross-check, riser DN80
[20] $h_{f,s2}$ = riser DN80: f (L + $L_{eq}$)/D · v²/2g + K v²/2g
= 0.020849 × (60 + 0) / 0.08 × 0.14014 + 8 × 0.14014
= 3.3123 m
Head loss, riser DN80
[21] $v_{s3}$ = index run DN65: Q / (π D²/4)
= 30 m³/h / (π × 65² mm² / 4)
= 2.5113 m/s
Velocity, index run DN65
[22] $Re_{s3}$ = index run DN65: ρ v D / μ
= 999.94 × 2.5113 × 0.065 / 0.0014684
= 111160
Reynolds number, index run DN65
[23] $f_{s3}$ = index run DN65: Darcy friction factor
= Colebrook–White, 9 iterations (Swamee–Jain start 0.020964)
= 0.020834
Friction factor, index run DN65
[24] $f_{sj,s3}$ = index run DN65: Swamee–Jain check
= 0.25 / log10(ε/3.7D + 5.74/Re^0.9)²
= 0.020964
Explicit cross-check, index run DN65
[25] $h_{f,s3}$ = index run DN65: f (L + $L_{eq}$)/D · v²/2g + K v²/2g
= 0.020834 × (45 + 6) / 0.065 × 0.32155 + 12 × 0.32155
= 9.1151 m
Head loss, index run DN65

[26] $h_{f,total} = \Sigma\ h_f$
= 3 segment(s)
= 12.696 m
Total friction and fitting loss at the design flow
[28] $V_{pipe} = \Sigma\ \pi/4 \cdot D^2 \cdot L$
= 3 segment(s)
= 513.75 L
Water held in the pipe segments
[29] $H_{static} = \Sigma\ dz + h_{out}$
= 12 + 3
= 15 m
Static head
[30] $H_{sys} = H_{static} + h_{f,total}$
= 15 m + 12.696 m
= 27.696 m
System head at the design flow
[31] $dp_{total} = \rho * 9.80665\ m/s^2 * h_{f,total}$
= 999.94 kg/m³ × 9.80665 m/s^2 × 12.696 m
= 124.5 kPa
Friction pressure drop at the design flow
[32] $dp_{sys} = \rho * 9.80665\ m/s^2 * H_{sys}$
= 999.94 kg/m³ × 9.80665 m/s^2 × 27.696 m
= 271.59 kPa
System pressure at the design flow
[33] $P_{hyd} = \rho * 9.80665\ m/s^2 * Q_{design} * H_{sys}$
= 999.94 kg/m³ × 9.80665 m/s^2 × 30 m³/h × 27.696 m
= 2.2633 kW
Hydraulic power at the design point
[40] $Q_{duty}$ = duty flow (as entered)
= $H_{pump}(Q) = H_{sys}(Q)$, scan and bisection
= 31.569 m³/h
Duty flow as entered
[41] $H_{duty}$ = duty head (as entered)
= $H_{sys}$(31.569 m³/h)
= 29.059 m
Duty head as entered
[42] $P_{shaft} = \rho\ g\ Q\ H\ /\ \eta$
= 999.94 × 9.8066 × 0.008769 × 29.059 / 0.68
= 3.6747 kW
Shaft power at duty as entered
[43] affinity = $Q_2 = Q_1\ (n_2/n_1)(D_2/D_1)$, $H_2 = H_1\ (n_2/n_1)^2(D_2/D_1)^2$ (impeller trim)
= $n_2/n_1 = 0.9$, $D_2/D_1 = 1$
= "curve scaled"
Affinity laws applied to the pump curve
[44] $Q_{duty2}$ = duty flow (after the affinity shift)
= $H_{pump}(Q) = H_{sys}(Q)$, scan and bisection
= 26.238 m³/h
Duty flow after the affinity shift
[45] $H_{duty2}$ = duty head (after the affinity shift)
= $H_{sys}$(26.238 m³/h)
= 24.712 m
Duty head after the affinity shift

[46] $P_{shaft2} = \rho\ g\ Q\ H\ /\ \eta$
= 999.94 × 9.8066 × 0.0072883 × 24.712 / 0.68
= 2.5972 kW
Shaft power at duty after the affinity shift
[49] $h_{f,suction}$ = Σ $h_f$ on the suction side
= 1 suction segment(s)
= 0.26906 m
Suction-side friction loss
[50] $NPSH_a = p_{atm} / (\rho * 9.80665\ m/s^2) + z_s - h_{f,suction} - p_v / (\rho * 9.80665\ m/s^2)$ (Ch. 8)
= 101.32 kPa / (999.94 kg/m³ × 9.80665 m/s^2) + 2 m − 0.26906 m − 0.93525 kPa / (999.94 kg/m³ × 9.80665 m/s^2)
= 11.968 m
NPSH available
[52] $u_{npsh} = NPSH_r / NPSH_a$
= 3 m / 11.968 m
= 0.25066
NPSH utilisation (required / available)
[53] $v_{peak}$ = highest segment velocity
= 2.5113 m/s
= 2.5113 m/s
Highest velocity
[54] $u_v = v_{peak} / v_{max}$
= 2.5113 / 2.5
= 1.0045
Velocity utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $Q_{design}$ | 30 m³/h | Design flow |
| $H_{static}$ | 15 m | Static head |
| $V_{pipe}$ | 513.7 L | Water held in the pipes |
| $H_{sys}$ | 27.70 m | System head at Q |
| $h_{f,total}$ | 12.70 m | Σ $h_f$ |
| $dp_{total}$ | 124.5 kPa | Friction Δp |
| $dp_{sys}$ | 271.6 kPa | System Δp |
| $P_{hyd}$ | 2.263 kW | Hydraulic power |
| $v_{peak}$ | 2.511 m/s | Highest velocity |
| $T_{line}$ | 6 °C | Line design temperature |
| $NPSH_a$ | 11.97 m | NPSH available |
| $Q_{duty}$ | 31.57 m³/h | Duty flow as entered |
| $H_{duty}$ | 29.06 m | Duty head as entered |
| $P_{shaft}$ | 3.675 kW | Shaft power as entered |
| $v_{s1}$ | 1.061 m/s | Velocity, plantroom suction |
| $Re_{s1}$ | 72253 | Re, plantroom suction |
| $f_{s1}$ | 0.021093 | f, plantroom suction |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $f_{sj,s1}$ | 0.021152 | f (Swamee–Jain), plantroom suction |
| $h_{f,s1}$ | 0.2691 m | $h_f$, plantroom suction |
| $v_{s2}$ | 1.658 m/s | Velocity, riser DN80 |
| $Re_{s2}$ | 90316 | Re, riser DN80 |
| $f_{s2}$ | 0.020849 | f, riser DN80 |
| $f_{sj,s2}$ | 0.020946 | f (Swamee–Jain), riser DN80 |
| $h_{f,s2}$ | 3.312 m | $h_f$, riser DN80 |
| $v_{s3}$ | 2.511 m/s | Velocity, index run DN65 |
| $Re_{s3}$ | 111160 | Re, index run DN65 |
| $f_{s3}$ | 0.020834 | f, index run DN65 |
| $f_{sj,s3}$ | 0.020964 | f (Swamee–Jain), index run DN65 |
| $h_{f,s3}$ | 9.115 m | $h_f$, index run DN65 |
| $Q_{duty2}$ | 26.24 m³/h | Duty flow after the affinity shift |
| $H_{duty2}$ | 24.71 m | Duty head after the affinity shift |
| $P_{shaft2}$ | 2.597 kW | Shaft power after the affinity shift |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| FAIL | Velocity ≤ limit | $u_v$ = 1.01 | $u_v$ ≤ 1 | 1.01 | |
| PASS | $NPSH_r \le NPSH_a$ | $u_{npsh}$ = 0.25 | $u_{npsh}$ ≤ 1 | 0.25 | |
| PASS | Duty point exists (as entered) | 31.569 m³/h | 29.059 m | – | |
| PASS | Duty point exists (after the affinity shift) | 26.238 m³/h | 24.712 m | – | |

## 7. Applicability

WARNING segments: index run DN65 runs at 2.5113 m/s, above the 2.5 m/s limit. [velocity]

Defaults used (inputs left empty):
circuit: not entered, so the default of 'open (the pump lifts the water by the rises entered)' is used.
$D_{law}$: not entered, so the default of 'impeller trim (Q ∝ n·D, H ∝ n²·D²)' is used.

## 8. Verification

```
Template bs.pipe-pump@1.9.0 Content hash 15a0bf02d74b6e79 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References


1. Crane Co. Flow of Fluids Through Valves, Fittings and Pipe. Technical Paper No. 410. Crane Co.
2. Çengel YA, Cimbala JM. Fluid Mechanics: Fundamentals and Applications. 4th ed. McGraw-Hill Education; 2018.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
4. Kell GS. Density, thermal expansivity, and compressibility of liquid water from 0° to 150°C: correlations and tables for atmospheric pressure and saturation reviewed and expressed on 1968 temperature scale. J Chem Eng Data. 1975;20(1):97-105.